

Heat up + Calibration Script

Explanation and Operating Logic

File analysed: 1. Heat up + Calibration_NEW TRY_v7.3 copia.py

Prepared for the ICN2 NPG chamber automation workflow

Purpose of this PDF: This document explains what the script does, how its logic is structured, how the GUI and hardware communication work, what safety protections are active, what files are saved, and how to operate the script in practice.

Section	What it explains
1	High-level purpose and role in the NPG synthesis sequence
2	Startup configuration, output folder and main default values
3	Hardware monitored or controlled by the script
4	Complete phase-by-phase workflow
5	Live GUI layout and operator controls
6	QMB calibration logic and final thickness ratio
7	Keysight current control, ramp-up modes and PID hold
8	Safety protections, watchdogs, abort and failsafe behaviour
9	Data logging, snapshots and final output files
10	Operator checklist, troubleshooting and function map

1. Executive summary

This script runs the first main experimental stage of the NPG chamber workflow: heating the CK-1 evaporator and performing a calibration deposition. It reads all relevant live signals, controls the Keysight power supply, regulates the CK-1 temperature using ramp-up logic plus a PID hold, guides the operator through shutter actions, and calculates the final calibration ratio used later by the DP-DBBA evaporation script.

- **Main experimental role:** heat the CK-1 evaporator until it is ready, open the shutter, deposit a known sample thickness, calculate CK-1 relative thickness / sample relative thickness, then ramp down safely.
- **Core automation:** it controls the Keysight current in remote mode, using either fixed current steps or a temperature-slope strategy until the CK-1 approaches the target temperature, then a PID hold keeps the CK-1 near the editable target.
- **Operator guidance:** the script waits for manual shutter confirmation in the GUI before starting calibration and again before ramp-down.
- **Critical output:** the final thickness ratio is the value that script 3 needs as input for DP-DBBA target-thickness calculation.

Most important result: The script calculates $\text{final_thickness_ratio} = \text{CK-1 relative thickness} / \text{Sample relative thickness}$. This is the calibration factor that links the CK-1 QMB reading to the sample deposition.

2. Where this script fits in the full workflow

Step	Script / action	Role
1	Heat up + Calibration - this script	Generates the CK-1 / sample thickness ratio by depositing until the sample relative thickness reaches 1.0 Å.
2	Sputtering-Annealing	Performs the sputtering and annealing preparation workflow using guided manual confirmations and live monitoring.
3	DP-DBBA Evaporation	Uses the ratio from this script to calculate the DP-DBBA CK-1 evaporation target.
4	NPG Annealings	Continues the process after evaporation and performs the final annealing plus current ramp-down sequence.

Connection with script 3: Script 3 asks for the thickness ratio obtained here. If this script is stopped before calibration finishes, that ratio may be missing or unreliable.

3. Startup configuration and output folder

At startup, the script asks for a run name. That name is sanitized and used to create an output folder next to the Python file.

```
sample_name = input("Enter the name for this evaporation run: ")
custom_folder_name = f"Heat up + Calibration data {safe_sample_name}"
final_folder_path = os.path.join(script_folder, custom_folder_name)
```

Item	Value / explanation
Output folder	Heat up + Calibration data <run name>
Data save interval	Every 5 seconds, the script writes updated text files with the current data arrays.
Live GUI refresh	Every 0.15 seconds, the Matplotlib interface is refreshed.
Initial phase	The process starts in HEATING_UP.

4. Main default values

Category	Parameter	Value in script	Meaning
Keysight current	KEYSIGHT_START_CURRENT_A	0.005 A	Small initial current used when remote automation starts.
Keysight current	KEYSIGHT_BASE_WORK_CURRENT_A	0.640 A	Upper working reference used by ramp logic.
Keysight current	KEYSIGHT_SOFT_WARNING_A	0.670 A	Normal cap: automation clamps current commands here.
Keysight current	KEYSIGHT_HARD_STOP_A	0.685 A	Software hard stop based on measured current or setpoint.

Category	Parameter	Value in script	Meaning
Keysight voltage	KEYSIGHT_VOLTAGE_LIMIT_V	2.30 V	Normal compliance voltage limit.
Keysight voltage	KEYSIGHT_HARD_STOP_V	2.45 V	Software hard stop based on measured voltage.
Heat-up target	HEATING_TRIGGER_TEMP_C	242.0 C	CK-1 temperature required before shutter opening can be requested.
Rate target	CK1_RATE_TARGET_A_PER_S	0.20 A/s	Average CK-1 QMB rate target required before shutter opening.
Calibration target	CALIBRATION_TARGET_SAMPLE_A	1.0 A	Sample relative thickness that ends the calibration deposition.
Ramp-down	RAMPDOWN_STEP_A	0.010 A	Current reduction step during safe ramp-down.
Ramp-down	RAMPDOWN_STEP_PERIOD_S	15 s	Time between ramp-down current steps.

5. Hardware interfaces used by the script

The script opens serial connections to all devices listed below. The port names and baud rates are hard-coded in the device_info dictionary.

Device key in code	Port	Baud	What the script does
CK-1 evaporator QMB	COM4	115200	Zeroes at startup; reads CK-1 thickness every 1 s and rate every 0.5 s.
Sample QMB	COM16	115200	Zeroes at startup; reads sample thickness every 1 s and rate every 0.5 s.
XGS600 HFIG pressure	COM6	9600	Reads synthesis chamber pressure using the #0002USYNTH command.
Oven PID temperature	COM9	9600	Reads oven PID process value M1 for monitoring only.
Keysight power supply	COM17	9600	Controls CK-1 crucible current and voltage compliance; reads voltage/current feedback.
Arduino CK-1 crucible temperature	COM3	9600	Reads the CK-1 temperature used for ramp control, PID hold and watchdog safety.

Important hardware note: The script assumes these COM ports are correct and available. If one device is disconnected or assigned to a different port, the corresponding monitoring or control thread may fail.

6. Complete phase-by-phase workflow

Phase	What happens	How it exits
HEATING_UP	Keysight remote automation starts. CK-1 current ramps up by the selected ramp strategy. When CK-1 is near target, PID hold starts regulating current around the live temperature target.	When CK-1 temperature is at or above the target and the average CK-1 rate is at or above the target rate.
WAIT_SHUTTER_OPEN	The script keeps PID regulation active and asks the operator to open the shutter and confirm it in the GUI or terminal.	Open Shutter confirmation sets the CK-1 and sample thickness baselines and starts CALIBRATION.
CALIBRATION	Relative thickness is calculated from the shutter-open baselines. The script waits until sample relative thickness reaches 1.0 A.	When sample relative thickness reaches the calibration target, the final ratio is calculated and the script asks to close the shutter.
WAIT_SHUTTER_CLOSE	The script waits for a fresh Close Shutter confirmation. It deliberately does not ramp down until this action is confirmed.	Close Shutter confirmation starts RAMP_DOWN.
RAMP_DOWN	Current is reduced by 0.010 A every 15 s until approximately 0 A. Output is switched OFF only at the end.	When ramp-down completes, phase becomes FINISHED.
FINISHED	Automation has ended normally. Final plots and saved data remain available.	The main loop stops or waits for plot windows depending on how the finish was requested.
SAFETY_STOP	Emergency or watchdog stop path. The Keysight is forced to 0 A and output OFF.	Used for abnormal stops and aborts.

Why shutter confirmation matters: The sample and CK-1 relative thickness values are only meaningful after the shutter

is opened. The script stores the QMB baselines when Open Shutter is confirmed.

7. Live GUI layout and operator controls

The interface is a Matplotlib window with 3 x 3 live plots on the left and a control panel on the right. The phase appears as a large title at the top.

GUI area	Controls / displays	Effect on the internal logic
Editable targets	Temperature target, target CK-1 rate, PID band, Apply and Reset buttons.	Updates live_heating_targets immediately. The heating condition, PID hold, watchdog and reference lines use the new values.
Ramp-up mode	Steps ramp, Slope ramp, steps-until temperature, step period, early/mid/late slope targets.	Controls how Keysight current increases before the target band is reached.
Manual Keysight current	Manual I, Set manual I, Back to Auto.	Manual mode pauses automatic Steps/Slope/PID current corrections but keeps safety checks and watchdog active.
Operator actions	Open Shutter, Close Shutter, Abort, Finish.	Controls phase progression and safe shutdown/ramp-down behaviour.
Run status	Phase, shutter status, current mode, ramp settings, targets, live CK-1 T/rate/current/voltage.	Gives the operator a compact view of the current state.
Last action	Shows the latest action or error message.	Helps confirm whether GUI clicks and target changes were accepted.

Live plot	Data shown
CK-1 thickness	Raw CK-1 QMB thickness and relative value once the shutter baseline exists.
CK-1 rate	CK-1 QMB rate, with target and band reference lines.
Pressure	XGS600 synthesis chamber pressure in mbar.
Sample thickness	Sample QMB thickness and relative value once the shutter baseline exists.
Sample rate	Sample QMB rate.
Oven temp	External oven PID process temperature.
Current	Keysight measured current.
Voltage	Keysight measured voltage.
CK-1 temp	Arduino CK-1 crucible temperature, with temperature target reference line.

8. QMB calibration logic and final thickness ratio

Both QMBs are zeroed at script startup. During the run, the script continuously reads their thickness and rate values. The calibration logic starts only after the operator confirms that the shutter is open.

```
def confirm_shutter_open(source='GUI button'):
    ck1_baseline = latest_ck1_thickness()
    sample_baseline = latest_sample_thickness()
    process_state['baseline_ck1_thickness'] = ck1_baseline
    process_state['baseline_sample_thickness'] = sample_baseline
    process_state['shutter_open_confirmed'] = True
```

Calculation	Code meaning	Purpose
relative_sample_thickness	latest sample thickness - sample baseline	Measures how much was deposited on the sample after opening the shutter.
relative_ck1_thickness	latest CK-1 thickness - CK-1 baseline	Measures how much the CK-1 QMB saw over the same open-shutter interval.
calibration end	sample relative thickness >= 1.0 A	Ends the calibration when the sample-side target is reached.
thickness ratio	CK-1 relative thickness / Sample relative thickness	Final calibration value to enter into the DP-DBBA evaporation script.

Interpretation of the ratio: If the ratio is high, the CK-1 QMB sees more thickness than the sample QMB for the same deposition event. Script 3 multiplies its desired sample-equivalent thickness by this ratio to obtain its CK-1 target.

9. Keysight current control and ramp-up modes

The Keysight power supply is configured in remote mode and used to control the CK-1 crucible coil current. The script separates normal operating limits from hard stops and from hardware latch protections.

Layer	Value / command	What it protects
Normal current cap	KEYSIGHT_SOFT_WARNING_A = 0.670 A	All normal current commands are clamped here.
Software hard current stop	KEYSIGHT_HARD_STOP_A = 0.685 A	If measured current or setpoint reaches this, the script stops the output.
Keysight OCP latch	0.690 A	Hardware latch set slightly above the software hard stop to avoid nuisance trips.
Voltage compliance limit	2.30 V	Normal voltage limit for the Keysight output.
Software hard voltage stop	2.45 V	If measured voltage reaches this, the script stops the output.
Keysight OVP latch	2.50 V	Hardware latch set above the software hard voltage stop.

9.1 Steps ramp-up mode

In the early part of heating, fixed current steps are safer and predictable. The script uses this mode until the CK-1 temperature reaches the editable steps-ramp threshold.

- Default threshold: 100.0 C.
- Default step size: +0.005 A.
- Default step period: 15 s.
- The current is still clamped by the normal soft cap.

9.2 Slope ramp-up mode

After the steps threshold, the script can regulate the current according to the measured CK-1 temperature slope. It estimates the temperature slope over recent Arduino readings and compares it with the selected early/mid/late target slope.

Current range	Target slope used	Default value
Current < 0.50 A	slope_early_c_per_min	9.0 C/min
0.50 A <= current < 0.60 A	slope_mid_c_per_min	8.0 C/min
Current >= 0.60 A	slope_late_c_per_min	7.0 C/min

9.3 Temperature PID hold

When the CK-1 temperature approaches the editable target band, the ramp-up strategy gives way to a PID temperature hold. This PID does not control the oven PID; it adjusts the Keysight current to stabilize the CK-1 crucible temperature.

PID parameter	Value	Meaning
PID_CONTROL_PERIOD_S	8.0 s	Minimum period between PID current corrections.
PID_TEMP_BAND_C	1.0 C	Inside this band, the PID holds current and does not chase noise.
PID_KP_A_PER_C	0.0020	Proportional current correction per temperature error.
PID_KI_A_PER_C_S	0.000030	Integral correction for accumulated error.
PID_KD_A_PER_C_PER_S	0.0030	Derivative correction for rapid error changes.
PID_MAX_STEP_A	0.0025 A	Maximum current change per PID correction.

Why the upper rate band is not blocking: The heating-to-open-shutter condition requires temperature >= target and average rate >= target. If the rate is above the old upper band, the script still allows progress instead of waiting unnecessarily.

10. Safety protections and stop behaviour

Protection	Trigger	Action
Soft current cap	A command requests current above 0.670 A.	Command is clamped to 0.670 A and a warning is printed.
Measured current hard stop	Measured current >= 0.685 A.	Keysight is forced to 0 A, output OFF, and stop_event is set.
Measured voltage hard stop	Measured voltage >= 2.45 V.	Keysight is forced to 0 A, output OFF, and stop_event is set.
Voltage compliance guard	Measured voltage reaches normal 2.30 V limit.	Current is reduced by one soft step without treating it as a hard-stop fault.

Protection	Trigger	Action
OCP/OVP latch detection	Keysight reports OCP/OVP trip or output OFF unexpectedly.	The script stops instead of silently continuing at 0 A.
Temperature watchdog soft action	CK-1 temperature $\geq$ target + 5 C.	PID is reset and current is reduced by up to 0.010 A.
Temperature watchdog hard stop	CK-1 temperature $\geq$ target + 10 C, stale sensor, impossible reading or unphysical jump.	Phase becomes SAFETY_STOP; Keysight current is forced to 0 A and output OFF.
Abort button / q command	Operator requests emergency/manual stop.	Immediate safe shutdown path forces Keysight zero and output OFF.
Python exit failsafe	Process exits before normal completion.	atexit handler calls emergency shutdown.

Normal finish versus Abort: A normal calibration finish waits for shutter close and performs a controlled ramp-down. Abort is immediate emergency-style shutdown and does not wait for the normal phase sequence.

11. Data logging and saved output files

The script saves data in the output folder created from the run name. It also saves phase summaries and graph-only snapshots at important transitions.

Saved item	When it is saved	Content
Device text files	Every 5 s during the main loop.	For each device, the script writes the time arrays and data arrays as text.
Phase summary text files	At heating end, shutter close, calibration end, ramp-down start, manual snapshots and safety stops.	Current phase, reason, latest values, relative thickness and thickness ratio if available.
Graph-only snapshots	At important phase transitions and manual snapshot requests.	Clean 3 x 3 measurement plots without GUI buttons or desktop capture.
Final Heat up + Calibration plot PNG	At program finalization.	Final 3 x 3 summary plot with QMB, pressure, oven, Keysight and CK-1 temperature data.
Console output	Continuously.	Live readings, status messages, warnings and final ratio.

Useful files to check after a run: The final plot confirms the full experiment visually. The phase summary around calibration_end or finished_after_rampdown is the easiest place to find the final thickness ratio.

12. Operator controls and terminal shortcuts

Action	GUI control	Terminal shortcut	Effect
Open shutter confirmation	Open Shutter	o	Stores QMB baselines and allows CALIBRATION to start.
Close shutter confirmation	Close Shutter	c	Allows safe ramp-down after calibration target is reached.
Save snapshot	No dedicated GUI button in this version	s	Requests a graph-only snapshot and saves a phase summary.
Show ratio	Shown in saved summaries / console after calculation	r	Prints CK-1 relative / Sample relative thickness ratio.
Show live targets	Visible in GUI	h	Prints the live heating target dictionary.
Stop / abort	Abort	q	Emergency shutdown; Keysight forced to zero and output OFF.
Controlled finish	Finish	No direct shortcut	Enters RAMP_DOWN from the GUI.

13. Practical run checklist

- Before running, confirm the correct COM ports and that all instruments are powered and connected.
- Run the Python file and enter a clear run name.
- Confirm the GUI opens and the live plots begin receiving data: QMBs, pressure, oven PID, Keysight and CK-1 temperature.
- During HEATING_UP, check that the CK-1 temperature and CK-1 rate are increasing as expected.
- When the script enters WAIT_SHUTTER_OPEN, manually open the shutter and click Open Shutter.
- During CALIBRATION, monitor the sample relative thickness. The target is 1.0 A.
- When TARGET REACHED appears, close the shutter physically and click Close Shutter.
- Allow the script to perform RAMP_DOWN. Do not switch off the Keysight before the controlled ramp-down reaches zero unless an emergency occurs.
- After the run, record the final thickness ratio for use in the DP-DBBA evaporation script.

14. Troubleshooting guide

Symptom	Likely cause	What to check
No live data appears for one device.	Wrong COM port, disconnected device, another program using the port, or serial parse failure.	Check Device Manager, cable, baud rate, and whether the instrument is already open elsewhere.
The script stays in HEATING_UP too long.	Temperature target or average CK-1 rate target has not been reached, or sensor data are missing.	Check CK-1 temperature plot, CK-1 rate plot, target values, and Arduino/QMB readings.
Current stops increasing at 0.670 A.	Soft current cap is active.	This is expected. The automation will not command above the normal cap.
Keysight output switches OFF unexpectedly.	Hardware OCP/OVP latch, output-off state, or hard stop condition.	Check console message for OCP/OVP/output-off diagnostic and measured I/V values.
Temperature watchdog hard stop occurs.	CK-1 is too hot, sensor stale, impossible reading, or sudden jump.	Check Arduino CK-1 temperature cable, live readings, target temperature, and whether the PID/current response was appropriate.
Ratio is None or missing.	Shutter baselines were not created, sample relative thickness is zero, or calibration did not reach target.	Confirm Open Shutter was clicked and sample relative thickness reached 1.0 A.
Final plot is empty or incomplete.	Run ended early, live data arrays were not filled, or device threads failed.	Check text data files and console errors for serial failures.

15. Function map for code maintenance

Function / block	Role in the script
monitor_qmb()	Zeroes and continuously reads CK-1 and Sample QMB thickness/rate.
read_pressure()	Reads synthesis chamber pressure from XGS600.
read_powersupply()	Reads Keysight current/voltage and performs voltage/current/protection checks.
read_PID()	Reads the external oven PID process temperature.
read_arduino()	Reads CK-1 crucible temperature from Arduino.
automate_keysight_heating()	Main automatic current-control loop: ramp-up, PID hold, manual-current pause and current caps.
apply_temperature_slope_control()	Adjusts current according to measured CK-1 temperature slope.
apply_temperature_pid_control()	Adjusts current to keep CK-1 temperature around the live target.
temperature_safety_watchdog()	Independent CK-1 temperature safety layer above the PID.
process_controller()	Phase machine: heating, shutter wait, calibration, close-shutter wait, ramp-down and finish.
confirm_shutter_open()	Stores QMB baselines and marks shutter open.
calculate_thickness_ratio()	Calculates CK-1 relative thickness divided by Sample relative thickness.
rampdown_keysight_output()	Controlled current ramp-down before output OFF.
save_phase_summary()	Saves compact text records of important transitions.
save_graphs_only_snapshot()	Saves clean measurement-only PNG snapshots.
_failsafe_on_exit()	Emergency shutdown if Python exits unexpectedly.

Maintenance recommendation: Because this script is a hardware-control script, changes should be limited and tested carefully. Avoid changing safety constants, phase transitions or current-control functions unless the effect is fully understood.